Karunya Mekala

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Summary

Full-Stack Developer and Computer Science graduate with 2+ years of professional experience building scalable web applications and data-driven systems. Pursuing an M.S. in Computer Science at the University of North Texas with a strong background in Python, Java, React, and machine learning. Seeking a Software Engineer or ML/Data Science role to deliver high-impact, production-ready solutions.

Technical Skills

Languages	Python, Java, JavaScript, C, C++, SQL, HTML/CSS
Frameworks	React, Node.js, Express, Flask, TensorFlow, Scikit-learn, Pandas, NumPy
ML / AI	Machine Learning, Deep Learning, NLP, Computer Vision, Time-Series Forecasting
ML Techniques	SMOTE, PCA, LASSO, Random Forest, XGBoost, LSTM, LightGBM, ARIMA/SARIMA
Cloud & Tools	AWS S3, Git/GitHub, MongoDB, SQL/NoSQL, REST APIs, Jupyter Notebook
Metrics & Eval	F1-Score, ROC-AUC, RMSE, Cross-Validation, Confusion Matrix

Experience

Full-Stack Developer | *Firenix Technologies • Pune, India* — Mar 2022 – Jun 2024

- Developed and maintained full-stack web applications using **React, Node.js, and Express**, handling everything from responsive UI components to RESTful API design and integration.
- Designed **MySQL and MongoDB schemas** for high-traffic workflows and implemented **JWT-based authentication** with role-based access control across client-facing applications.
- Worked in an **Agile/Scrum** team — code reviews, sprint planning, and third-party API integrations — consistently delivering features on schedule.

Projects

Student Academic Performance & Dropout Risk Predictor | Python • XGBoost • Random Forest • SMOTE • Neural Networks — 2024

- Built a predictive classification system using Random Forest, XGBoost, and Neural Networks on demographic, academic, and LMS engagement data to flag at-risk students before traditional methods detect failure.
- Addressed class imbalance with SMOTE and evaluated models via F1-Score and ROC-AUC, producing a decision-support tool that helps educators intervene early and improve retention outcomes.

BCI Fault-Based Repair Framework | Python • Signal Processing • Test Oracles • Fault Detection — 2024

- Engineered a fault repair framework for Brain-Computer Interface systems using partial test oracles, corrective heuristics, and slice functions — achieving **100% Out-of-Bounds** and **94% Multimodal Inconsistency** detection rates.
- Validated across five BCI applications; delivered a **26% reduction in system faults** while preserving overall model accuracy.

Full-Stack Restaurant Management Platform | Node.js • React • MongoDB • AWS S3 • Socket.IO — 2024

- Engineered a MERN-stack web application with live order status updates via Socket.IO, role-based JWT authentication, and cloud media storage on AWS S3 — deployed end-to-end on Render/Vercel.

Education

M.S. Computer Science — In Progress
University of North Texas — Denton, TX — *GPA: 3.5 / 4.0*

B.Tech. Computer Science & Engineering — Apr 2024
Jain University — Bangalore, India — *CGPA: 8.42 / 10*

Certifications

- Object-Oriented Programming with Java *(LinkedIn)*
- Diploma in Computer Applications *(Wave Infotech)*
- Java 17 Masterclass *(Udemy)*
- Chegg Expert Learning Platform *(Chegg)*